



Long division

Division is something that we use all the time, and you won't always have a calculator at hand. So it's worth learning how to divide large numbers with just a pen and paper.

Short division

Short division means dividing a large number by a one-digit number.

Write $651 \div 3$
like this:

$$3 \overline{)651}$$

- 1) Divide each digit of the large number by the one-digit number, from left to right.
- 2) If you get a remainder, put this in front of the next digit along.

This isn't as hard as I thought.



How many 3s in 6? 2.

$$3 \overline{)651} \quad \begin{matrix} 2 \\ \hline \end{matrix}$$

(You're really dividing 600 by 3.)

How many 3s in 5? 1, with a remainder of 2.

$$3 \overline{)651} \quad \begin{matrix} 21 \\ \hline \end{matrix} \quad \text{Remainder}$$

(You're actually dividing 50 by 3.)

How many 3s in 21? 7.

$$3 \overline{)651} \quad \begin{matrix} 217 \\ \hline \end{matrix}$$

Now have a go!

$$7 \overline{)826}$$

$$6 \overline{)468}$$

$$2 \overline{)472}$$

$$7 \overline{)616}$$

$$8 \overline{)568}$$

TOP TIP

Another way to work out division problems is to use multiplication and guesswork. This is called trial and error.

Imagine you're trying to divide 108 by 3.

$40 \times 3 = 120$. This is too big.

$30 \times 3 = 90$. This is too small.

$35 \times 3 = 105$. Nearly there...

$36 \times 3 = 108$.

What happens if you try to divide a number by zero?

It doesn't matter how many zeros you add together – they will never add up to a whole number. You could go on adding $0 + 0 + 0 + 0$ forever.

This is why any number divided by zero equals **infinity**.